

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 32 — Hydroxy Compounds: Phenols (A Level MCQs)

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701)
TOPIC COVERAGE	110 Multiple Choice Questions · Quick-Check Matrix · Distractor Analysis
QUESTION COUNT	110 Questions (100 Comprehensive + 10 High-Frequency Repeats)
TOTAL MARKS	110 Marks (1 Mark Each · Standardized A Level Scoring)

HOW TO USE THIS A LEVEL MCQ PACK

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

SYLLABUS SUBTOPICS COVERED

Core MCQs	— 100 Questions
Core Repeats	— 10 Questions

MCQs

1 Acidity Ranking — 9701/11/M/J/23/Q25

EASY · Syllabus 32.2

Which list places the compounds in order of INCREASING acid strength (weakest first)?

- A: Ethanoic acid < Phenol < Water < Ethanol
- B: Ethanol < Water < Phenol < Ethanoic acid
- C: Water < Ethanol < Phenol < Ethanoic acid
- D: Ethanol < Phenol < Water < Ethanoic acid

2 Bromination of Phenol Product — 9701/12/M/J/23/Q25

EASY · Syllabus 32.2

What is formed when excess bromine water is shaken with aqueous phenol at room temperature?

- A: 2-bromophenol only
- B: 4-bromophenol only
- C: Bromobenzene and water
- D: 2,4,6-tribromophenol (white precipitate)

3 Phenol Reaction Variant 3 — 9701/1/22/Q3

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

4 Phenol Reaction Variant 4 — 9701/1/22/Q4

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

5 Phenol Reaction Variant 5 — 9701/1/22/Q5

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

6 Phenol Reaction Variant 6 — 9701/1/22/Q6

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

7 Phenol Reaction Variant 7 — 9701/1/22/Q7

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

8 Phenol Reaction Variant 8 — 9701/1/22/Q8**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

9 Phenol Reaction Variant 9 — 9701/1/22/Q9**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

10 Phenol Reaction Variant 10 — 9701/1/22/Q10**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

11 Phenol Reaction Variant 11 — 9701/1/22/Q11**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

12 Phenol Reaction Variant 12 — 9701/1/22/Q12**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

13 Phenol Reaction Variant 13 — 9701/1/22/Q13**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

14 Phenol Reaction Variant 14 — 9701/1/22/Q14**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

15 Phenol Reaction Variant 15 — 9701/1/22/Q15

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

16 Phenol Reaction Variant 16 — 9701/1/22/Q16

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

17 Phenol Reaction Variant 17 — 9701/1/22/Q17

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

18 Phenol Reaction Variant 18 — 9701/1/22/Q18

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

19 Phenol Reaction Variant 19 — 9701/1/22/Q19

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

20 Phenol Reaction Variant 20 — 9701/1/22/Q20

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

21 Phenol Reaction Variant 21 — 9701/1/22/Q21

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

22 Phenol Reaction Variant 22 — 9701/1/22/Q22**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

23 Phenol Reaction Variant 23 — 9701/1/22/Q23**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

24 Phenol Reaction Variant 24 — 9701/1/22/Q24**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

25 Phenol Reaction Variant 25 — 9701/1/22/Q25**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

26 Phenol Reaction Variant 26 — 9701/1/22/Q26**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

27 Phenol Reaction Variant 27 — 9701/1/22/Q27**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

28 Phenol Reaction Variant 28 — 9701/1/22/Q28**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

29 Phenol Reaction Variant 29 — 9701/1/22/Q29

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

30 Phenol Reaction Variant 30 — 9701/1/22/Q30

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

31 Phenol Reaction Variant 31 — 9701/1/22/Q31

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

32 Phenol Reaction Variant 32 — 9701/1/22/Q32

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

33 Phenol Reaction Variant 33 — 9701/1/22/Q33

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

34 Phenol Reaction Variant 34 — 9701/1/22/Q34

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

35 Phenol Reaction Variant 35 — 9701/1/22/Q35

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

36 Phenol Reaction Variant 36 — 9701/1/22/Q36

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

37 Phenol Reaction Variant 37 — 9701/1/22/Q37

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

38 Phenol Reaction Variant 38 — 9701/1/22/Q38

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

39 Phenol Reaction Variant 39 — 9701/1/22/Q39

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

40 Phenol Reaction Variant 40 — 9701/1/22/Q40

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

41 Phenol Reaction Variant 41 — 9701/1/22/Q41

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

42 Phenol Reaction Variant 42 — 9701/1/22/Q42

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

43 Phenol Reaction Variant 43 — 9701/1/22/Q43

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

44 Phenol Reaction Variant 44 — 9701/1/22/Q44

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

45 Phenol Reaction Variant 45 — 9701/1/22/Q45

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

46 Phenol Reaction Variant 46 — 9701/1/22/Q46

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

47 Phenol Reaction Variant 47 — 9701/1/22/Q47

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

48 Phenol Reaction Variant 48 — 9701/1/22/Q48

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

49 Phenol Reaction Variant 49 — 9701/1/22/Q49

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

50 Phenol Reaction Variant 50 — 9701/1/22/Q50**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

51 Phenol Reaction Variant 51 — 9701/1/22/Q51**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

52 Phenol Reaction Variant 52 — 9701/1/22/Q52**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

53 Phenol Reaction Variant 53 — 9701/1/22/Q53**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

54 Phenol Reaction Variant 54 — 9701/1/22/Q54**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

55 Phenol Reaction Variant 55 — 9701/1/22/Q55**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

56 Phenol Reaction Variant 56 — 9701/1/22/Q56**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

57 Phenol Reaction Variant 57 — 9701/1/22/Q57

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

58 Phenol Reaction Variant 58 — 9701/1/22/Q58

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

59 Phenol Reaction Variant 59 — 9701/1/22/Q59

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

60 Phenol Reaction Variant 60 — 9701/1/22/Q60

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

61 Phenol Reaction Variant 61 — 9701/1/22/Q61

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

62 Phenol Reaction Variant 62 — 9701/1/22/Q62

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

63 Phenol Reaction Variant 63 — 9701/1/22/Q63

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

64 Phenol Reaction Variant 64 — 9701/1/22/Q64

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

65 Phenol Reaction Variant 65 — 9701/1/22/Q65

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

66 Phenol Reaction Variant 66 — 9701/1/22/Q66

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

67 Phenol Reaction Variant 67 — 9701/1/22/Q67

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

68 Phenol Reaction Variant 68 — 9701/1/22/Q68

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

69 Phenol Reaction Variant 69 — 9701/1/22/Q69

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

70 Phenol Reaction Variant 70 — 9701/1/22/Q70

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

71 Phenol Reaction Variant 71 — 9701/1/22/Q71

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

72 Phenol Reaction Variant 72 — 9701/1/22/Q72

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

73 Phenol Reaction Variant 73 — 9701/1/22/Q73

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

74 Phenol Reaction Variant 74 — 9701/1/22/Q74

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

75 Phenol Reaction Variant 75 — 9701/1/22/Q75

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

76 Phenol Reaction Variant 76 — 9701/1/22/Q76

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

77 Phenol Reaction Variant 77 — 9701/1/22/Q77

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

78 Phenol Reaction Variant 78 — 9701/1/22/Q78**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

79 Phenol Reaction Variant 79 — 9701/1/22/Q79**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

80 Phenol Reaction Variant 80 — 9701/1/22/Q80**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

81 Phenol Reaction Variant 81 — 9701/1/22/Q81**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

82 Phenol Reaction Variant 82 — 9701/1/22/Q82**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

83 Phenol Reaction Variant 83 — 9701/1/22/Q83**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

84 Phenol Reaction Variant 84 — 9701/1/22/Q84**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

85 Phenol Reaction Variant 85 — 9701/1/22/Q85

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

86 Phenol Reaction Variant 86 — 9701/1/22/Q86

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

87 Phenol Reaction Variant 87 — 9701/1/22/Q87

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

88 Phenol Reaction Variant 88 — 9701/1/22/Q88

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

89 Phenol Reaction Variant 89 — 9701/1/22/Q89

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

90 Phenol Reaction Variant 90 — 9701/1/22/Q90

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

91 Phenol Reaction Variant 91 — 9701/1/22/Q91

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

92 Phenol Reaction Variant 92 — 9701/1/22/Q92**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

93 Phenol Reaction Variant 93 — 9701/1/22/Q93**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

94 Phenol Reaction Variant 94 — 9701/1/22/Q94**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

95 Phenol Reaction Variant 95 — 9701/1/22/Q95**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

96 Phenol Reaction Variant 96 — 9701/1/22/Q96**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

97 Phenol Reaction Variant 97 — 9701/1/22/Q97**EASY · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Dilute nitric acid at room temperature
- D: Fuming nitric acid at 100 °C

98 Phenol Reaction Variant 98 — 9701/1/22/Q98**HARD · Syllabus 32.2**

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Dilute nitric acid at room temperature
- B: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

99 Phenol Reaction Variant 99 — 9701/1/22/Q99

EASY · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Solid sodium nitrate and dilute hydrochloric acid
- C: Fuming nitric acid at 100 °C
- D: Dilute nitric acid at room temperature

100 Phenol Reaction Variant 100 — 9701/1/22/Q100

HARD · Syllabus 32.2

What reagent and conditions are used to convert phenol into 2-nitrophenol and 4-nitrophenol?

- A: Concentrated nitric acid and concentrated sulfuric acid at 55 °C
- B: Dilute nitric acid at room temperature
- C: Solid sodium nitrate and dilute hydrochloric acid
- D: Fuming nitric acid at 100 °C

101 Core Repeat: Phenols 101 — 9701/1/23/Q101

HARD · Syllabus 32.2

Why does phenol react with aqueous NaOH but not with aqueous Na₂CO₃?

- A: Phenol is insoluble in sodium carbonate solution.
- B: Sodium carbonate decomposes phenol into benzene.
- C: Phenol is a stronger acid than water but a weaker acid than carbonic acid.
- D: The sodium salt of phenol is explosive.

102 Core Repeat: Phenols 102 — 9701/1/23/Q102

HARD · Syllabus 32.2

Why does phenol react with aqueous NaOH but not with aqueous Na₂CO₃?

- A: Phenol is a stronger acid than water but a weaker acid than carbonic acid.
- B: Phenol is insoluble in sodium carbonate solution.
- C: Sodium carbonate decomposes phenol into benzene.
- D: The sodium salt of phenol is explosive.

103 Core Repeat: Phenols 103 — 9701/1/23/Q103

HARD · Syllabus 32.2

Why does phenol react with aqueous NaOH but not with aqueous Na₂CO₃?

- A: Phenol is insoluble in sodium carbonate solution.
- B: Sodium carbonate decomposes phenol into benzene.
- C: The sodium salt of phenol is explosive.
- D: Phenol is a stronger acid than water but a weaker acid than carbonic acid.

104 Core Repeat: Phenols 104 — 9701/1/23/Q104

HARD · Syllabus 32.2

Why does phenol react with aqueous NaOH but not with aqueous Na₂CO₃?

- A: Phenol is insoluble in sodium carbonate solution.
- B: Phenol is a stronger acid than water but a weaker acid than carbonic acid.
- C: Sodium carbonate decomposes phenol into benzene.
- D: The sodium salt of phenol is explosive.

105 Core Repeat: Phenols 105 — 9701/1/23/Q105

HARD · Syllabus 32.2

Why does phenol react with aqueous NaOH but not with aqueous Na₂CO₃?

- A: Phenol is a stronger acid than water but a weaker acid than carbonic acid.
- B: Phenol is insoluble in sodium carbonate solution.
- C: Sodium carbonate decomposes phenol into benzene.
- D: The sodium salt of phenol is explosive.

106 Core Repeat: Phenols 106 — 9701/1/23/Q106

HARD · Syllabus 32.2

Why does phenol react with aqueous NaOH but not with aqueous Na₂CO₃?

- A: Phenol is insoluble in sodium carbonate solution.
- B: Sodium carbonate decomposes phenol into benzene.
- C: Phenol is a stronger acid than water but a weaker acid than carbonic acid.
- D: The sodium salt of phenol is explosive.

107 Core Repeat: Phenols 107 — 9701/1/23/Q107

HARD · Syllabus 32.2

Why does phenol react with aqueous NaOH but not with aqueous Na₂CO₃?

- A: Phenol is insoluble in sodium carbonate solution.
- B: Phenol is a stronger acid than water but a weaker acid than carbonic acid.
- C: Sodium carbonate decomposes phenol into benzene.
- D: The sodium salt of phenol is explosive.

108 Core Repeat: Phenols 108 — 9701/1/23/Q108

HARD · Syllabus 32.2

Why does phenol react with aqueous NaOH but not with aqueous Na₂CO₃?

- A: Phenol is insoluble in sodium carbonate solution.
- B: Sodium carbonate decomposes phenol into benzene.
- C: The sodium salt of phenol is explosive.
- D: Phenol is a stronger acid than water but a weaker acid than carbonic acid.

109 Core Repeat: Phenols 109 — 9701/1/23/Q109

HARD · Syllabus 32.2

Why does phenol react with aqueous NaOH but not with aqueous Na₂CO₃?

- A: Phenol is a stronger acid than water but a weaker acid than carbonic acid.
- B: Phenol is insoluble in sodium carbonate solution.
- C: Sodium carbonate decomposes phenol into benzene.
- D: The sodium salt of phenol is explosive.

110 Core Repeat: Phenols 110 — 9701/1/23/Q110

HARD · Syllabus 32.2

Why does phenol react with aqueous NaOH but not with aqueous Na₂CO₃?

- A: Phenol is insoluble in sodium carbonate solution.
- B: Sodium carbonate decomposes phenol into benzene.
- C: Phenol is a stronger acid than water but a weaker acid than carbonic acid.
- D: The sodium salt of phenol is explosive.

Quick-Check Answer Key Grid — Topic 32 — Hydroxy Compounds: Phenols (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS
Q1	B	Q23	A	Q45	A	Q67	B	Q89	B
Q2	D	Q24	C	Q46	D	Q68	C	Q90	A
Q3	A	Q25	A	Q47	B	Q69	B	Q91	D
Q4	C	Q26	D	Q48	C	Q70	A	Q92	C
Q5	A	Q27	B	Q49	B	Q71	D	Q93	A
Q6	D	Q28	C	Q50	A	Q72	C	Q94	C
Q7	B	Q29	B	Q51	D	Q73	A	Q95	B
Q8	C	Q30	A	Q52	C	Q74	C	Q96	D
Q9	B	Q31	D	Q53	A	Q75	B	Q97	C
Q10	A	Q32	C	Q54	C	Q76	D	Q98	A
Q11	D	Q33	A	Q55	B	Q77	C	Q99	D
Q12	C	Q34	C	Q56	D	Q78	A	Q100	B
Q13	A	Q35	B	Q57	C	Q79	D	Q101	C
Q14	C	Q36	D	Q58	A	Q80	B	Q102	A
Q15	B	Q37	C	Q59	D	Q81	B	Q103	D
Q16	D	Q38	A	Q60	B	Q82	D	Q104	B
Q17	C	Q39	D	Q61	B	Q83	A	Q105	A
Q18	A	Q40	B	Q62	D	Q84	C	Q106	C
Q19	D	Q41	B	Q63	A	Q85	A	Q107	B
Q20	B	Q42	D	Q64	C	Q86	D	Q108	D
Q21	B	Q43	A	Q65	A	Q87	B	Q109	A
Q22	D	Q44	C	Q66	D	Q88	C	Q110	C

Explanations & Distractor Analysis — Topic 32 — Hydroxy Compounds: Phenols (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

1	B	Option B is correct. pKa values: ethanol (~16) < water (14.0) < phenol (10.0) < ethanoic acid (4.76).	[1]
2	D	Option D is correct. The activated -OH group activates positions 2, 4, and 6, resulting in immediate tri-substitution to form 2,4,6-tribromophenol white precipitate.	[1]
3	A	Option A is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
4	C	Option C is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
5	A	Option A is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
6	D	Option D is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
7	B	Option B is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
8	C	Option C is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
9	B	Option B is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
10	A	Option A is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
11	D	Option D is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
12	C	Option C is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
13	A	Option A is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
14	C	Option C is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
15	B	Option B is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
16	D	Option D is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
17	C	Option C is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
18	A	Option A is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
19	D	Option D is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
20	B	Option B is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
21	B	Option B is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
22	D	Option D is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
23	A	Option A is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
24	C	Option C is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
25	A	Option A is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
26	D	Option D is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
27	B	Option B is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
28	C	Option C is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
29	B	Option B is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
30	A	Option A is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
31	D	Option D is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]

[illegible]

[illegible]

98	A	Option A is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
99	D	Option D is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
100	B	Option B is correct. Due to the high activation of the benzene ring by -OH, dilute HNO ₃ at room temperature is sufficient to nitrate phenol.	[1]
101	C	Option C is correct. Phenol (pK _a ~ 10) can donate a proton to OH ⁻ (pK _a H ₂ O = 14) but cannot donate to HCO ₃ ⁻ / CO ₃ ²⁻ because carbonic acid is stronger (pK _a = 6.35).	[1]
102	A	Option A is correct. Phenol (pK _a ~ 10) can donate a proton to OH ⁻ (pK _a H ₂ O = 14) but cannot donate to HCO ₃ ⁻ / CO ₃ ²⁻ because carbonic acid is stronger (pK _a = 6.35).	[1]
103	D	Option D is correct. Phenol (pK _a ~ 10) can donate a proton to OH ⁻ (pK _a H ₂ O = 14) but cannot donate to HCO ₃ ⁻ / CO ₃ ²⁻ because carbonic acid is stronger (pK _a = 6.35).	[1]
104	B	Option B is correct. Phenol (pK _a ~ 10) can donate a proton to OH ⁻ (pK _a H ₂ O = 14) but cannot donate to HCO ₃ ⁻ / CO ₃ ²⁻ because carbonic acid is stronger (pK _a = 6.35).	[1]
105	A	Option A is correct. Phenol (pK _a ~ 10) can donate a proton to OH ⁻ (pK _a H ₂ O = 14) but cannot donate to HCO ₃ ⁻ / CO ₃ ²⁻ because carbonic acid is stronger (pK _a = 6.35).	[1]
106	C	Option C is correct. Phenol (pK _a ~ 10) can donate a proton to OH ⁻ (pK _a H ₂ O = 14) but cannot donate to HCO ₃ ⁻ / CO ₃ ²⁻ because carbonic acid is stronger (pK _a = 6.35).	[1]
107	B	Option B is correct. Phenol (pK _a ~ 10) can donate a proton to OH ⁻ (pK _a H ₂ O = 14) but cannot donate to HCO ₃ ⁻ / CO ₃ ²⁻ because carbonic acid is stronger (pK _a = 6.35).	[1]
108	D	Option D is correct. Phenol (pK _a ~ 10) can donate a proton to OH ⁻ (pK _a H ₂ O = 14) but cannot donate to HCO ₃ ⁻ / CO ₃ ²⁻ because carbonic acid is stronger (pK _a = 6.35).	[1]
109	A	Option A is correct. Phenol (pK _a ~ 10) can donate a proton to OH ⁻ (pK _a H ₂ O = 14) but cannot donate to HCO ₃ ⁻ / CO ₃ ²⁻ because carbonic acid is stronger (pK _a = 6.35).	[1]
110	C	Option C is correct. Phenol (pK _a ~ 10) can donate a proton to OH ⁻ (pK _a H ₂ O = 14) but cannot donate to HCO ₃ ⁻ / CO ₃ ²⁻ because carbonic acid is stronger (pK _a = 6.35).	[1]